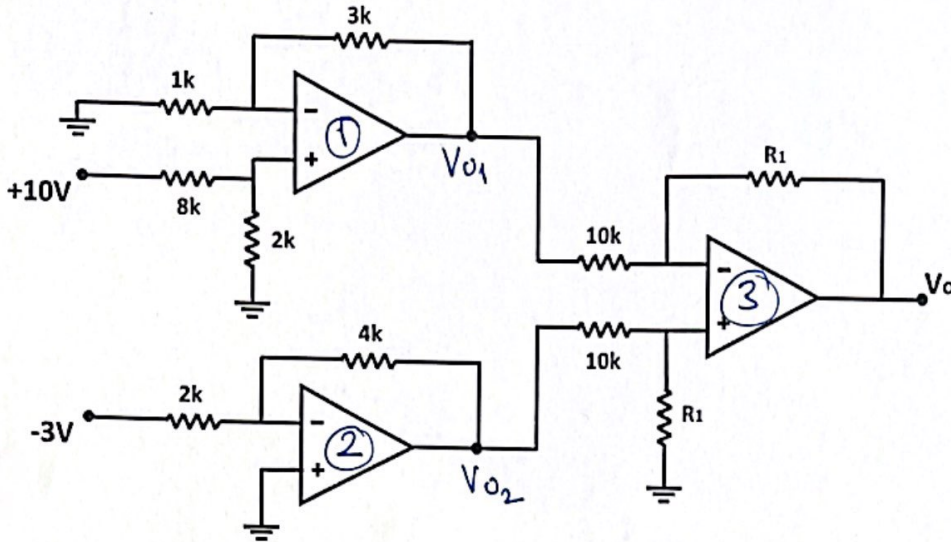




$V_o = -4V$ ise $R_1 = ?$

Çözüm

① nolu op-amp (testeneyen yükseltici)

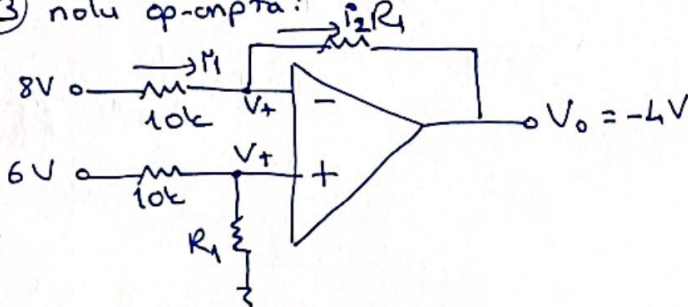
$$V_{o1} = \left(\frac{3k}{1k} + 1 \right) \cdot V_+ \leftarrow V_+ = \frac{10V}{8k+2k} \cdot 2k = 2V$$

$$\boxed{V_{o1} = 8V} \quad (8 \text{ puan})$$

② nolu op-amp (testeneyen yükseltici)

$$V_{o2} = - \frac{4k}{2k} \cdot (-3) \Rightarrow \boxed{V_{o2} = 6V} \quad (8 \text{ puan})$$

③ nolu op-amp'ta:



$$V_+ = \frac{6}{10k+R_1} \cdot R_1 = V_-$$

$$i_1 = i_2$$

$$\frac{8 - \frac{6R_1}{10k+R_1}}{10k} = \frac{\frac{6R_1}{10k+R_1} - (-4)}{R_1}$$

$$\frac{8R_1 - 6R_1^2}{10k+R_1} = \frac{60kR_1}{10k+R_1} + 40k$$

$$80kR_1 + 8R_1^2 - 6R_1^2 = 60kR_1 + 400k^2 + 40kR_1$$

$$2R_1^2 + 20kR_1 - 400k^2 = 0$$

$$R_1^2 + 10kR_1 - 200k^2 = 0$$

$$\begin{matrix} -20 & +10 \end{matrix}$$

$$\boxed{R_1 = 20k\Omega}$$

ve ya;

çıkarma devresi:

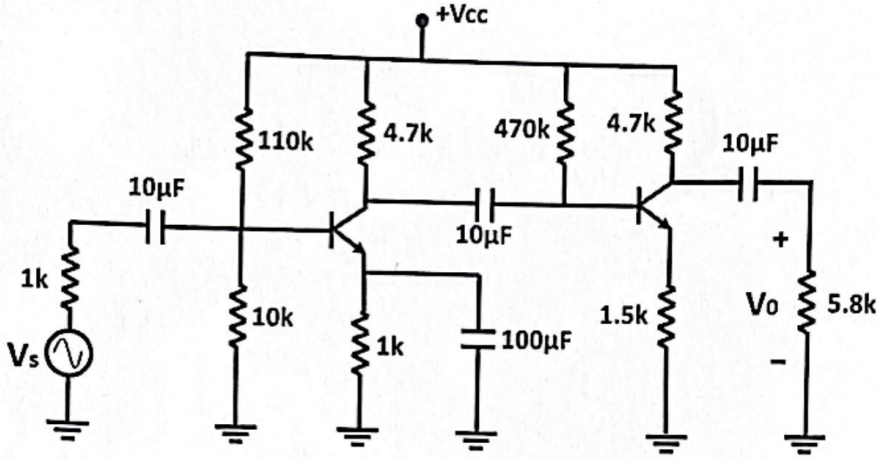
$$V_o = \frac{R_1}{10k} (V_{o2} - V_{o1})$$

$$-4 = \frac{R_1}{10k} (6 - 8)$$

$$\frac{2}{10k} = \frac{R_1}{10k} \Rightarrow R_1 = 20k\Omega$$

$$\boxed{R_1 = 20k\Omega}$$

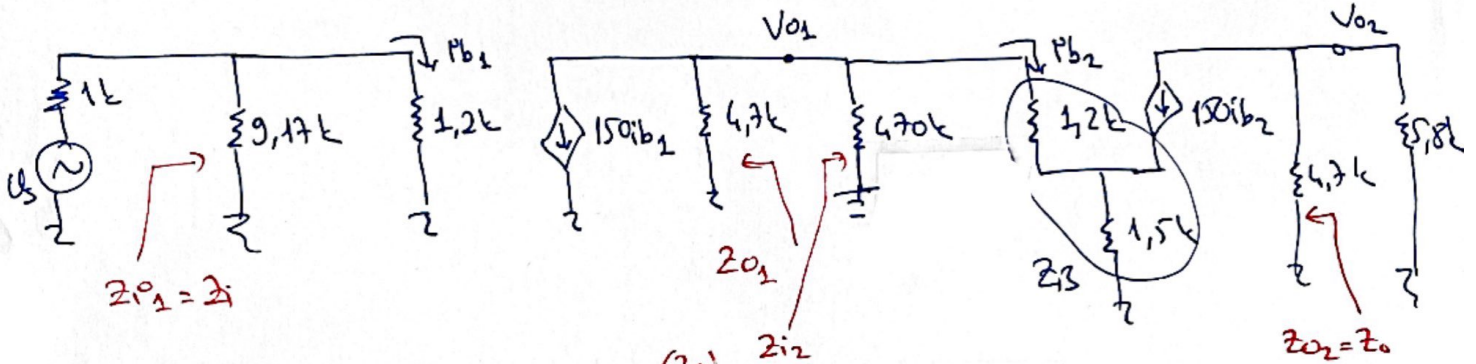
(9 puan)



Yandaki yükselteç devresi için
her iki transistöre ait
parametreler şu şekildedir:
 $h_{ie1}=h_{ie2}=1.2k$; $h_{fe1}=h_{fe2}=150$

Buna göre;

- $Z_{i1}, Z_{i2}, Z_{o1}, Z_{o2}, Z_i, Z_o = ?$
- $A_{v1}, A_{v2}, A_v = ?$
- $A_{vs} = ?, A_i = ?$



a) $Z_i = Z_{i1} = 9.17k \parallel 1.2k$

$Z_p = Z_{p1} = 1.06k\Omega$ (2p)

$Z_{o1} = 4.7k$ (2p)
 $Z_{o2} = 4.7k$ (2p)

$Z_o = 4.7k$ (2p)

$Z_{i2} = 470k \parallel (1.2k + 150 \cdot 1.5k)$

$Z_{i2} = 470k \parallel 227.7k$

$Z_{i2} \approx 153.4k\Omega$ (2p)

b) $A_{v1} = \frac{V_{o1}}{V_{i1}} = - \frac{150i_{b1}(4.7k \parallel Z_{i2})}{i_{b1} \cdot 1.2k} \Rightarrow A_{v1} = -570$ (3p)

$A_{v2} = \frac{V_{o2}}{V_{i2}} = - \frac{150i_{b2}(4.7k \parallel 5.8k)}{i_{b2} \cdot 227.7k} \Rightarrow A_{v2} = -1.71$ (3p)

$A_v = A_{v1} \cdot A_{v2} \Rightarrow A_v = -570 \cdot -1.71 \Rightarrow A_v = 974.7$ (2p)

c) $A_{vs} = A_v \cdot \frac{Z_i}{Z_i + R_s} = 974.7 \cdot \frac{0.98k}{0.98k + 1k} \Rightarrow A_{vs} = 482.42$ (3p)

$A_i = -A_v \cdot \frac{Z_i}{R_L} = -974.7 \cdot \frac{0.98k}{5.8k} \approx -164.7 \Rightarrow A_i = -164.7$ (2p)